Supplementary Material: Auditing Physics Constraints in Wind Power Forecasting

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Overview

This supplementary document provides complete supporting ledgers, empirical evaluation logs, inter-rater reliability data, and reporting compliance tables for the main manuscript:

- **Section S1 (Table S1)**: The complete study-level scoping retrieval ledger and multi-dimensional coding table across all 48 evaluated records (46 included studies and 2 explicitly excluded studies).
- **Section S2 (Table S2)**: Detailed inter-rater agreement statistics between Coder A and independent Coder C (Juntao Du) across the calibration subset ($N = 26$).
- **Section S3 (Table S3)**: Full numerical results for the 3-Arm Guardrail benchmark across all 48 experimental configurations (5 random seeds $\times$ 4 noise regimes $\times$ 6 penalty weights).
- **Section S4 (Table S4)**: Exploratory Kelmarsh gradient-matching downstream optimization logs across 5 independent seeds.
- **Section S5 (Table S5)**: PRISMA-ScR scoping review checklist compliance table.

Section S1: Complete Scoping Audit Corpus Ledger

Table 1: Comprehensive 48-Study Scoping Audit Ledger and Lifecycle Classifications (Schema v4).

ID	Year	Study Title	D1: Provenance	D2: Stage	D3: Telemetry	D4: Fidelity	D5: Ablation
S01	2025	Physics-constrained wind power forecasting al...	historical-static	training-loss	none	not-found	constraint-on-off
S02	2026	Physics-constrained transformer for wind powe...	historical-static	training-loss	none	code-mismatch	constraint-on-off
S03	2026	Physics-constrained generative augmentation a...	intrinsic	data-generation	none	not-found	module-ablation
S04	2026	A Physics-Constrained Dual-Stream Dynamic Fra...	intrinsic	training-loss	forecast-input	not-found	both
S05	2026	A physics-constrained temporal modeling frame...	intrinsic	training-loss	none	not-found	constraint-on-off
S06	2026	Physics-Constrained and Generalization-Enhanc...	intrinsic	training-loss	none	not-found	constraint-on-off
S07	2026	NCF: A physics-constrained pre-trained method...	intrinsic	inference-architecture	forecast-input	not-found	module-ablation
S08	2026	Physics-constrained graph attention learning ...	intrinsic	training-loss	none	not-found	both
S09	2024	Physics-informed reinforcement learning for p...	intrinsic	training-loss	none	not-found	constraint-on-off
S10	2025	Prediction of wind turbines power with physic...	intrinsic	training-loss	none	code-mismatch	constraint-on-off
S11	2026	A novel non-dimensional multiple operating co...	intrinsic	training-loss	none	not-found	constraint-on-off
S12	2025	Physics-Informed Adaptive-Weight NBeatsx for ...	intrinsic	inference-architecture	none	not-found	module-ablation
S13	2025	High-precision short-term predicting method o...	intrinsic	training-loss	none	not-found	both
S14	2025	Physics-Informed Neural Networks for Enhanced...	intrinsic	training-loss	none	not-found	constraint-on-off
S15	2026	PSTG-Net: A physics-informed spatiotemporal m...	intrinsic	inference-architecture	none	not-found	module-ablation
S16	2026	Physics informed virtual sensor modeling for ...	intrinsic	training-loss	none	not-found	constraint-on-off
S17	2025	Physics informed data-driven approach for ult...	intrinsic	inference-architecture	forecast-input	not-found	module-ablation
S18	2026	Physics-informed dynamic wake graphs for prob...	intrinsic	inference-architecture	none	not-found	both
S19	2026	Physics-Informed Adaptive Conformal Predictio...	intrinsic	post-hoc-correction	none	not-found	constraint-on-off
S20	2026	A branch-adaptive framework for multi-timesca...	intrinsic	inference-architecture	none	not-found	module-ablation
S21	2026	Physics-informed LLM-guided dynamic spatio-te...	intrinsic	inference-architecture	none	not-found	module-ablation
S22	2026	Physics-informed neural network modelling of ...	intrinsic	training-loss	none	not-found	constraint-on-off
S23	2026	A Physics-informed Operating Condition Attent...	intrinsic	inference-architecture	none	not-found	module-ablation
S24	2025	Physics-Informed Hybrid KNN-MLP for Enhanced ...	intrinsic	inference-architecture	none	not-found	module-ablation
S25	2026	Physics-Informed dual-branch graph neural net...	intrinsic	training-loss	none	not-found	both
S26	2026	*Excluded: wind-speed-only / no-explicit-power-constraint*					
S27	2026	UniWind: Toward Unified Day-Ahead Wind Power ...	historical-static	inference-architecture	forecast-input	not-found	module-ablation

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Table 1 – *Continued from previous page*

ID	Year	Study Title	D1: Provenance	D2: Stage	D3: Telemetry	D4: Fidelity	D5: Ablation
S28	2026	Platform-Yaw-Aware Physics-Informed Meta-Lear...	intrinsic	training-loss	none	not-found	constraint-on-off
S29	2026	A Dual-Track Feature-Enhanced Physics-Informe...	intrinsic	training-loss	none	not-found	constraint-on-off
S30	2026	DSTC-Net: A Physics-Informed Dynamic Spatio-T...	intrinsic	inference-architecture	none	not-found	module-ablation
S31	2026	Short-Term Solar-Wind Power Forecasting via P...	intrinsic	inference-architecture	none	not-found	module-ablation
S32	2026	Adaptive Physics-Informed Neural Network for ...	learned-online	inference-architecture	none	not-found	module-ablation
S33	2026	High-Penetration New Energy Power System Outa...	intrinsic	training-loss	contemporaneous-state	not-found	constraint-on-off
S34	2023	Physics-Guided Machine Learning for Wind-Farm...	intrinsic	inference-architecture	none	not-found	module-ablation
S35	2026	Development of a physics-guided bidirectional...	intrinsic	training-loss	none	not-found	constraint-on-off
S36	2026	Spatiotemporal physics-guided graph neural ne...	intrinsic	inference-architecture	none	not-found	module-ablation
S37	2025	Regional wind power forecasting guided by phy...	intrinsic	inference-architecture	none	not-found	module-ablation
S38	2025	Wind Power Prediction Method Based on Physics...	historical-static	training-loss	none	not-found	constraint-on-off
S39	2026	Physics-Guided Deep Learning for Short-Term O...	intrinsic	inference-architecture	forecast-input	not-found	module-ablation
S40	2026	Physics-guided multiscale fusion network for ...	intrinsic	inference-architecture	none	not-found	module-ablation
S41	2026	Physics-Guided Spatio-Temporal Transformers t...	intrinsic	training-loss	none	not-found	constraint-on-off
S42	2026	*Excluded: no-explicit-physics-constraint*					
S43	2025	Physics-Guided Short-Term Offshore Wind Power...	intrinsic	inference-architecture	none	not-found	module-ablation
S44	2026	Short-term wind power forecasting for data-li...	intrinsic	training-loss	none	not-found	both
S45	2026	Cross-Domain Few-Shot Wind Power Forecasting ...	historical-static	inference-architecture	none	not-found	module-ablation
S46	2026	Enhancing Wind Power Forecasting via Multi-Fa...	intrinsic	inference-architecture	contemporaneous-state	not-found	module-ablation
S47	2026	Advancing short-term wind power forecasting v...	intrinsic	inference-architecture	forecast-input	not-found	module-ablation
S48	2026	Physics-guided tri-modal spatio-temporal grap...	intrinsic	inference-architecture	none	not-found	module-ablation

Section S2: Inter-Rater Reliability and Arbitration on Representative Sample ($N = 20$)

Table 2 documents the double-blind inter-rater reliability metrics between the primary coder (Coder A) and the independent second coder (Coder C) across the stratified representative sample ($N = 20$, comprising 18 eligible studies and 2 negative controls). All evaluations were conducted on physically isolated coding packages without preset answers.

Table 2: Comprehensive Inter-Rater Reliability Metrics on the Representative Sample ($N = 20$).

Audited Dimension	N	Observed P_o	Chance P_e	Cohen's κ	95% CI	Gwet's AC_1	Concordance Level
Eligibility (D0)	20	100.0%	82.0%	**1.000**	[1.000, 1.000]	**1.000**	Almost Perfect (S26, S42 excluded)
D1: Provenance	18	100.0%	65.4%	**1.000**	[1.000, 1.000]	**1.000**	Almost Perfect
D2: Stage	18	94.4%	40.1%	**0.907**	[0.730, 1.000]	**0.931**	Almost Perfect
D3: Telemetry	18	88.9%	64.8%	**0.684**	[0.272, 1.000]	**0.830**	Substantial
D4: Code Fidelity	18	94.4%	84.6%	**0.640**	[-0.046, 1.000]	**0.934**	Substantial (Kappa Paradox, AC_1 =
D5: Ablation Type	18	83.3%	37.7%	**0.733**	[0.457, 1.000]	**0.759**	Substantial
D5: Harm Reporting	18	100.0%	100.0%	**1.000**	[1.000, 1.000]	**1.000**	Almost Perfect (18/18 zero harm re

Table 3 itemizes all identified coding discrepancies and documents the definitive arbitration decisions and textual evidence.

Table 3: Adjudicated Discrepancies and Formal Arbitration Rationale for the Sampled Literature.

ID	Dimension	Coder A	Coder C	Final Decision	Arbitration Rationale and Ground Truth Evidence
S10	D4: Code Fidelity	`not-found`	`code-mismatch`	`code-mismatch`	Coder C independently retrieved and verified the official repository (`alfonsogijon/WindTurbines_PINNs`): code implements static algebraic `sn.Tie` ($P = gT\omega$), omitting the paper's claimed ODE residual $J\dot{\omega} = T_a - T_g$, λ_{ode}, and evidential NLL, with normalization divisor bug (`.mean()` instead of `.std()`).
S18	D2: Stage	`training-loss`	`inference-arch.`	`inference-arch.`	Jensen wake deficit equations are hard-coded into dynamic graph adjacency topology during forward inference; loss penalty (Eq. 11) is statistical quantile non-crossing, not a physical equation.
S19	D5: Ablation Type	`module-ablation`	`constraint-on-off`	`constraint-on-off`	Conformal prediction base architecture is identical; Table 3 strictly tests activating vs. deactivating the post-hoc physical boundary clamping.
S26	D4 / D5	`not-found / no`	`n/a / n/a`	`n/a`	Both coders agreed on exclusion under D0 (forecasts wind speed only); downstream dimensions are standardly N/A.
S27	D3: Telemetry	`none`	`forecast-input`	`forecast-input`	Section 4 explicitly states day-ahead forecasting is driven by future NWP forecast wind fields.
S27	D5: Ablation Type	`both`	`module-ablation`	`module-ablation`	Table 5 excises the entire State Routing module; does not isolate physical prior from network parameters.
S34	D5: Ablation Type	`both`	`module-ablation`	`module-ablation`	Control baseline is a standalone ML model completely lacking the Gaussian wake feature extraction block.
S39	D3: Telemetry	`none`	`forecast-input`	`forecast-input`	First-stage physical aerodynamic conversion requires future WRF forecast wind speeds to compute baseline power.
S42	D4 / D5	`not-found / no`	`n/a / n/a`	`n/a`	Both coders agreed on exclusion under D0 (heuristic feature partitioning, no physical equation).

Section S3: Extended 3-Arm Guardrail Benchmark Results

Table 4 presents the complete numerical results for the 3-Arm Guardrail ablation across all 48 evaluated experimental configurations. Each metric reports the sample mean ± standard deviation across 5 independent random seeds (42, 123, 456, 789, 1024).

Table 4: Complete Downstream Guardrail Evaluation across Noise Regimes and Penalty Weights (Wind-CN, 5 Seeds).

Experimental Strategy / Arm	σ	λ	Post-hoc Clip	RMSE (kW)	MAE (kW)	Violation Rate (%)
Arm 1: No Guardrail	0.0	0.00	No	6151.3 ± 48.7	3907.4 ± 39.1	1.14% ± 0.31%
Arm 1: No Guardrail	0.2	0.00	No	15148.7 ± 82.3	11773.4 ± 76.9	6.81% ± 0.65%
Arm 1: No Guardrail	0.4	0.00	No	28428.6 ± 119.6	22254.2 ± 100.2	16.52% ± 0.53%
Arm 1: No Guardrail	0.6	0.00	No	41923.5 ± 432.7	32919.9 ± 312.9	25.76% ± 0.59%
Arm 2: Post-hoc Clipping	0.0	0.00	Yes	6148.1 ± 50.9	3937.6 ± 35.4	0.00% ± 0.00%
Arm 2: Post-hoc Clipping	0.2	0.00	Yes	14522.8 ± 30.7	11262.6 ± 21.8	0.00% ± 0.00%
Arm 2: Post-hoc Clipping	0.4	0.00	Yes	25007.6 ± 161.7	19529.8 ± 131.2	0.00% ± 0.00%
Arm 2: Post-hoc Clipping	0.6	0.00	Yes	33553.8 ± 174.4	26333.1 ± 133.0	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.0	0.01	Yes	6148.4 ± 50.8	3937.9 ± 35.5	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.0	0.05	Yes	6149.6 ± 50.3	3939.3 ± 35.8	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.0	0.10	Yes	6151.0 ± 49.7	3941.0 ± 36.2	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.0	0.50	Yes	6162.3 ± 46.0	3954.6 ± 39.6	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.0	1.00	Yes	6175.1 ± 43.3	3969.1 ± 44.7	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.2	0.01	Yes	14520.2 ± 30.8	11260.6 ± 21.8	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.2	0.05	Yes	14509.9 ± 30.9	11252.6 ± 22.0	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.2	0.10	Yes	14497.0 ± 31.1	11242.7 ± 22.3	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.2	0.50	Yes	14394.9 ± 33.8	11163.9 ± 25.4	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.2	1.00	Yes	14270.1 ± 38.3	11067.3 ± 30.1	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.4	0.01	Yes	25002.9 ± 161.4	19526.3 ± 130.9	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.4	0.05	Yes	24984.1 ± 160.0	19512.1 ± 129.9	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.4	0.10	Yes	24960.7 ± 158.2	19494.4 ± 128.6	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.4	0.50	Yes	24775.2 ± 144.7	19354.2 ± 118.6	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.4	1.00	Yes	24547.8 ± 128.7	19181.9 ± 106.6	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.6	0.01	Yes	33548.3 ± 174.3	26328.9 ± 132.9	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.6	0.05	Yes	33526.3 ± 173.9	26311.8 ± 132.6	0.00% ± 0.00%

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Table 4 – *Continued from previous page*

Experimental Strategy / Arm	σ	λ	Post-hoc Clip	RMSE (kW)	MAE (kW)	Violation Rate (%)
Arm 3: Joint Guardrail (Clipped)	0.6	0.10	Yes	33498.9 ± 173.4	26290.4 ± 132.3	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.6	0.50	Yes	33281.1 ± 171.2	26121.0 ± 130.5	0.00% ± 0.00%
Arm 3: Joint Guardrail (Clipped)	0.6	1.00	Yes	33011.7 ± 170.4	25911.3 ± 129.8	0.00% ± 0.00%
Arm 3: Joint Guardrail (Raw)	0.0	0.01	No	6151.5 ± 48.6	3907.8 ± 39.2	1.14% ± 0.31%
Arm 3: Joint Guardrail (Raw)	0.0	0.05	No	6152.5 ± 48.1	3909.1 ± 39.5	1.13% ± 0.30%
Arm 3: Joint Guardrail (Raw)	0.0	0.10	No	6153.8 ± 47.5	3910.8 ± 40.0	1.13% ± 0.30%
Arm 3: Joint Guardrail (Raw)	0.0	0.50	No	6163.3 ± 43.6	3924.1 ± 43.8	1.09% ± 0.24%
Arm 3: Joint Guardrail (Raw)	0.0	1.00	No	6174.5 ± 40.9	3938.5 ± 49.1	1.06% ± 0.20%
Arm 3: Joint Guardrail (Raw)	0.2	0.01	No	15145.5 ± 82.3	11770.9 ± 76.9	6.81% ± 0.65%
Arm 3: Joint Guardrail (Raw)	0.2	0.05	No	15132.8 ± 82.6	11761.2 ± 77.1	6.79% ± 0.65%
Arm 3: Joint Guardrail (Raw)	0.2	0.10	No	15116.9 ± 82.9	11749.1 ± 77.3	6.77% ± 0.65%
Arm 3: Joint Guardrail (Raw)	0.2	0.50	No	14992.3 ± 85.4	11653.5 ± 79.0	6.65% ± 0.63%
Arm 3: Joint Guardrail (Raw)	0.2	1.00	No	14841.4 ± 88.5	11537.2 ± 81.1	6.50% ± 0.61%
Arm 3: Joint Guardrail (Raw)	0.4	0.01	No	28421.6 ± 119.4	22248.9 ± 100.0	16.52% ± 0.53%
Arm 3: Joint Guardrail (Raw)	0.4	0.05	No	28393.4 ± 118.6	22227.5 ± 99.3	16.50% ± 0.53%
Arm 3: Joint Guardrail (Raw)	0.4	0.10	No	28358.3 ± 117.7	22201.0 ± 98.5	16.47% ± 0.54%
Arm 3: Joint Guardrail (Raw)	0.4	0.50	No	28082.3 ± 110.9	21991.6 ± 92.4	16.26% ± 0.53%
Arm 3: Joint Guardrail (Raw)	0.4	1.00	No	27748.2 ± 103.8	21737.2 ± 85.6	16.00% ± 0.53%
Arm 3: Joint Guardrail (Raw)	0.6	0.01	No	41912.7 ± 432.7	32911.7 ± 313.0	25.76% ± 0.59%
Arm 3: Joint Guardrail (Raw)	0.6	0.05	No	41869.7 ± 432.9	32879.0 ± 313.2	25.73% ± 0.59%
Arm 3: Joint Guardrail (Raw)	0.6	0.10	No	41816.0 ± 433.0	32838.4 ± 313.4	25.69% ± 0.59%
Arm 3: Joint Guardrail (Raw)	0.6	0.50	No	41394.2 ± 433.8	32518.1 ± 314.7	25.40% ± 0.60%
Arm 3: Joint Guardrail (Raw)	0.6	1.00	No	40883.4 ± 433.5	32128.8 ± 315.6	25.05% ± 0.60%

Section S4: Downstream Kelmarsh Optimization Benchmark

Table 5 details the 5-seed exploratory gradient-matching optimization experiment on the open Kelmarsh SCADA dataset (40 epochs per seed), verifying the downstream convergence consequences of divergence argument directionality.

Table 5: Downstream Convergence Comparison on Kelmarsh Wind Farm Telemetry (Test RMSE across 5 Seeds).

Objective Formulation	**Seed 0**	**Seed 1**	**Seed 2**	**Seed 3**	**Seed 4**	**Mean ± Std**
Unconstrained MSE Baseline	0.0835	0.0789	0.0800	0.0694	0.0780	**0.0780 ± 0.0052**
Paper-Specified Symmetric JS	0.1485	0.1194	0.1016	0.0948	0.1116	0.1152 ± 0.0209
Codebase Reverse-JS Divergence	0.2134	0.1462	0.2675	0.2883	0.2459	0.2323 ± 0.0555
Direct Uncalibrated 1:1 Reverse KL	0.5388	0.4223	0.4075	0.5173	0.4647	0.4701 ± 0.0574

Section S5: PRISMA-ScR Scoping Review Checklist Compliance

Table 6 documents adherence to the 22 reporting items of the PRISMA-ScR statement.

Table 6: PRISMA-ScR Itemized Scoping Review Reporting Checklist.

Item No.	**PRISMA-ScR Checklist Item Description**	**Manuscript Location**
1	Title: Identify the report as a scoping review/audit	Title, Abstract
2	Structured summary: Objectives, eligibility, participants, results, conclusions	Abstract
3	Rationale: Describe the rationale for the review	Section 1 (Introduction)
4	Objectives: Provide an explicit statement of questions being addressed	Section 1.1, Section 8.1
5	Protocol and registration: Indicate whether a protocol exists	Section 4
6	Eligibility criteria: Specify characteristics used as criteria for inclusion	Section 4.2
7	Information sources: Describe all information sources in the search	Section 4.1
8	Search: Present the full electronic search strategy for at least one database	Section 4.1
9	Selection of sources of evidence: Process for selecting sources	Section 4.2, Figure 2
10	Data charting process: Methods of charting data	Section 4.3
11	Data items: List and define all variables for which data were sought	Section 3.2, Table 2
12	Critical appraisal: Auditing methodology and fidelity	Section 4, Section 5
13	Synthesis of results: Summarize the scoping review findings	Section 5, Table 3, Table 4
14	Selection of sources of evidence: Report results of search and screening	Section 4.2, Figure 2
15	Characteristics of sources of evidence: Present data on study characteristics	Section 5.1, Table 3
16	Critical appraisal of individual sources: Summarize code/ablation audit	Section 5.2, Section 5.3
17	Results of individual sources of evidence: Detailed case study	Section 6 (PCT case)
18	Synthesis of results: Summarize quantitative lifecycle distributions	Section 5.1, Figure 3
19	Summary of evidence: Summarize main results and implications	Section 8.1
20	Limitations: Discuss limitations of the scoping review process	Section 9 (Limitations)
21	Conclusions: Provide a general interpretation of the results	Section 10 (Conclusion)
22	Funding: Describe sources of funding and competing interests	Frontmatter, End matter